

## WAUKEGAN NATIONAL, IL, USA

WMO: 725347

Lat: 42.417N

Lon: 87.867W

Elev: 727

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 14880

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.4       | 2.1 | -12.2                         | 2.9 | -1.3 | -8.0 | 3.7 | 3.4  | 25.2                  | 35.7 | 23.1 | 32.8 | 7.2                   | 270  | 0.535 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.4                   | 89.5            | 74.2 | 86.0 | 72.6 | 82.9 | 71.0 | 76.5                | 85.7 | 74.7 | 82.8 | 73.1 | 80.5 | 11.4                 | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.3                            | 127.2 | 81.6 | 72.1 | 122.1 | 79.9 | 70.5 | 115.3 | 77.8 | 40.4          | 85.7 | 38.6 | 82.9 | 37.1 | 80.5 | 82.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| 22.8              | 19.4 | 17.4 | DB                         | -8.6 | 93.9               | 5.9 | 3.7                                                | -12.9 | 96.5       | -16.3 | 98.6       | -19.6 | 100.7      | -23.9 | 103.3 | (4) |
|                   |      |      | WB                         | -9.3 | 79.3               | 5.7 | 1.6                                                | -13.4 | 80.4       | -16.8 | 81.4       | -20.0 | 82.3       | -24.1 | 83.5  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 48.3  | 23.3  | 25.3  | 35.4  | 46.1  | 56.2  | 66.1  | 71.5  | 70.5  | 63.7  | 51.8  | 40.0  | 28.4  | (6)  |
| (7)  |                                                                   | DBStd       | 19.22 | 11.48 | 11.03 | 10.79 | 9.03  | 8.87  | 7.40  | 6.33  | 5.47  | 7.64  | 8.67  | 9.63  | 10.71 | (7)  |
| (8)  |                                                                   | HDD50       | 3285  | 827   | 692   | 472   | 179   | 35    | 0     | 0     | 0     | 2     | 84    | 323   | 671   | (8)  |
| (9)  |                                                                   | HDD65       | 6711  | 1291  | 1110  | 918   | 572   | 301   | 74    | 15    | 12    | 115   | 418   | 751   | 1134  | (9)  |
| (10) |                                                                   | CDD50       | 2675  | 0     | 1     | 19    | 63    | 229   | 485   | 667   | 634   | 413   | 140   | 22    | 2     | (10) |
| (11) |                                                                   | CDD65       | 627   | 0     | 0     | 1     | 5     | 30    | 108   | 217   | 181   | 76    | 9     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 4919  | 0     | 0     | 9     | 73    | 272   | 949   | 1800  | 1236  | 523   | 57    | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 1413  | 0     | 0     | 1     | 14    | 57    | 288   | 581   | 336   | 128   | 8     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 8.0   | 9.0   | 8.9   | 8.9   | 9.6   | 8.2   | 6.9   | 6.3   | 6.0   | 6.7   | 8.0   | 8.7   | 8.7   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 35.9  | 1.9   | 1.6   | 2.2   | 3.8   | 4.0   | 4.1   | 3.5   | 3.9   | 3.2   | 3.0   | 2.7   | 2.0   | (15) |
| (16) |                                                                   | PrecMax     | 44.5  | 3.7   | 4.6   | 4.4   | 7.6   | 8.4   | 9.7   | 7.2   | 11.3  | 9.5   | 6.9   | 5.9   | 4.5   | (16) |
| (17) |                                                                   | PrecMin     | 23.6  | 0.5   | 0.2   | 0.7   | 1.7   | 0.6   | 1.0   | 1.3   | 1.9   | 0.7   | 0.7   | 0.4   | 0.8   | (17) |
| (18) |                                                                   | PrecStd     | 4.8   | 0.9   | 1.0   | 1.1   | 1.6   | 1.8   | 2.2   | 1.6   | 1.9   | 1.7   | 1.8   | 1.5   | 1.0   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 55.5  | 60.6  | 73.2  | 81.7  | 85.4  | 91.5  | 95.3  | 92.1  | 88.0  | 80.9  | 68.6  | 59.0  | (19) |
| (20) |                                                                   | MCWB        | 50.5  | 52.1  | 60.9  | 63.4  | 69.4  | 73.8  | 76.3  | 75.0  | 72.7  | 68.3  | 58.6  | 54.9  | (20)  |      |
| (21) |                                                                   | 2%          | DB    | 47.9  | 52.2  | 63.5  | 74.4  | 81.0  | 87.7  | 89.7  | 87.6  | 83.6  | 74.3  | 63.1  | 52.0  | (21) |
| (22) |                                                                   | MCWB        | 42.9  | 46.4  | 55.0  | 59.8  | 67.0  | 72.0  | 75.2  | 74.1  | 71.4  | 64.1  | 54.3  | 48.7  | (22)  |      |
| (23) |                                                                   | 5%          | DB    | 42.1  | 44.9  | 57.4  | 68.3  | 76.7  | 82.7  | 86.0  | 83.5  | 80.6  | 70.3  | 59.0  | 46.2  | (23) |
| (24) |                                                                   | MCWB        | 38.2  | 39.7  | 50.0  | 57.2  | 64.9  | 70.7  | 73.6  | 72.0  | 68.3  | 61.8  | 52.3  | 42.9  | (24)  |      |
| (25) |                                                                   | 10%         | DB    | 38.1  | 39.3  | 51.1  | 62.3  | 72.3  | 79.4  | 82.7  | 80.8  | 76.2  | 65.8  | 54.5  | 42.7  | (25) |
| (26) |                                                                   | MCWB        | 35.7  | 35.9  | 45.1  | 52.4  | 62.2  | 68.1  | 71.8  | 70.7  | 67.0  | 58.6  | 48.6  | 39.6  | (26)  |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 52.4  | 54.6  | 62.5  | 66.2  | 72.6  | 77.4  | 79.3  | 77.9  | 75.5  | 70.8  | 60.3  | 55.0  | (27) |
| (28) |                                                                   | MDCB        | 54.5  | 58.1  | 72.4  | 78.2  | 82.5  | 86.5  | 89.2  | 88.3  | 84.3  | 78.2  | 65.0  | 59.1  | (28)  |      |
| (29) |                                                                   | 2%          | WB    | 43.7  | 46.6  | 57.3  | 61.6  | 68.9  | 74.0  | 77.2  | 75.4  | 73.1  | 66.7  | 56.7  | 48.3  | (29) |
| (30) |                                                                   | MDCB        | 47.3  | 51.8  | 62.9  | 71.6  | 77.7  | 83.6  | 86.9  | 83.4  | 80.3  | 71.5  | 61.9  | 50.6  | (30)  |      |
| (31) |                                                                   | 5%          | WB    | 38.5  | 40.2  | 51.1  | 58.1  | 66.3  | 72.0  | 75.2  | 73.8  | 70.9  | 63.1  | 52.9  | 43.8  | (31) |
| (32) |                                                                   | MDCB        | 41.2  | 44.1  | 55.9  | 67.4  | 74.1  | 81.1  | 83.6  | 80.8  | 77.3  | 68.3  | 57.6  | 46.1  | (32)  |      |
| (33) |                                                                   | 10%         | WB    | 35.8  | 36.2  | 45.0  | 53.1  | 63.7  | 69.8  | 73.5  | 72.2  | 68.7  | 59.8  | 49.2  | 39.6  | (33) |
| (34) |                                                                   | MDCB        | 38.3  | 39.2  | 50.4  | 61.6  | 70.7  | 77.7  | 80.8  | 78.5  | 74.5  | 65.2  | 54.4  | 42.1  | (34)  |      |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 13.6  | 14.8  | 15.6  | 17.8  | 18.5  | 18.1  | 17.4  | 16.5  | 17.4  | 16.2  | 14.8  | 12.9  | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 17.4  | 19.5  | 24.3  | 28.5  | 24.7  | 22.2  | 20.2  | 20.0  | 21.1  | 21.7  | 20.0  | 16.1  | (36) |
| (37) |                                                                   | MCWBR       | 14.4  | 14.7  | 17.0  | 17.4  | 13.9  | 11.3  | 9.3   | 9.2   | 10.6  | 13.3  | 14.2  | 13.9  | (37)  |      |
| (38) |                                                                   | 5% WB       | MCDBR | 15.7  | 18.8  | 22.4  | 26.1  | 22.4  | 20.3  | 18.3  | 16.8  | 18.1  | 19.1  | 18.1  | 14.9  | (38) |
| (39) |                                                                   | MCWBR       | 13.6  | 15.0  | 17.2  | 17.6  | 13.9  | 11.7  | 10.1  | 8.9   | 10.3  | 13.7  | 14.7  | 14.0  | (39)  |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.292 | 0.308 | 0.341 | 0.394 | 0.419 | 0.436 | 0.436 | 0.429 | 0.400 | 0.350 | 0.311 | 0.291 | (40)  |      |
| (41) |                                                                   | taud        | 2.446 | 2.401 | 2.400 | 2.280 | 2.261 | 2.263 | 2.262 | 2.281 | 2.334 | 2.473 | 2.523 | 2.485 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 272   | 284   | 287   | 277   | 272   | 267   | 266   | 263   | 263   | 266   | 263   | 262   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 24    | 29    | 33    | 41    | 43    | 43    | 42    | 40    | 35    | 27    | 22    | 21    | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 472   | 752   | 1125  | 1459  | 1674  | 1919  | 1944  | 1684  | 1372  | 889   | 556   | 391   | (44)  |      |
| (45) |                                                                   | RadStd      | 61    | 91    | 93    | 120   | 131   | 127   | 90    | 103   | 117   | 101   | 56    | 43    | (45)  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (56 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page